Critical Slowing Down

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Abstract

This project explores critical slowing down in the two-dimensional Ising model near the critical temperature T_c. Using Markov chain Monte Carlo methods, we implement both the Metropolis and Wolff algorithms to generate spin configurations and measure autocorrelation times of magnetization and energy. Magnetic susceptibility is determined with an external field B, and, in the absence of the field, via the fluctuation–dissipation theorem (FDT). The scaling of relaxation times allows estimation of the dynamic critical exponent z for each algorithm. Comparisons between local and cluster updates highlight their relative efficiency at criticality, showing how algorithmic choices affect the sampling of Markov chains in systems exhibiting critical slowing down and field-induced responses.

1 Introduction: Dynamic Critical Exponents and Autocorrelation Times

Understanding dynamic properties of Monte Carlo simulations requires treating the update dynamics as a stochastic process. In this framework, the time evolution of a Monte Carlo configuration is described by a Markov chain. This provides the natural foundation for defining autocorrelation functions, relaxation times, and ultimately the dynamic critical exponent z.

1.1 Markov Chains and Master Equations

For relaxational models, such as the stochastic Ising model, the time-dependent behavior is described by a master equation [1]

$$\frac{\partial P_n(t)}{\partial t} = -\sum_{n \neq m} \left[P_n(t)W_{n\to m} - P_m(t)W_{m\to n}\right] \tag{1}$$

where $P_n(t)$ is the probability of the system being in state n at time t, and $W_{n\to m}$ is the transition rate for $n \to m$. The solution to the master equation is a sequence of states, but the time variable is a stochastic quantity which does not represent true time. A relaxation function $\phi(t)$ can be defined which describes time correlations *within equilibrium*

$$\phi_{MM}(t) = \frac{\langle M(0)M(t)\rangle - \langle M\rangle^2}{\langle M^2\rangle - \langle M\rangle^2}. \tag{2}$$

When normalized in this way, the relaxation function is 1 at $t = 0$ and decays to zero as $t \to \infty$. It is important to remember that for a system in equilibrium any time in the sequence of states may be chosen as the '$t = 0$' state. The asymptotic, long time behavior of the relaxation function is exponential, i.e.

$$\phi(t) \to e^{-t/\tau}. \tag{3}$$

where the correlation time τ diverges as T_c is approached. This dynamic (relaxational) critical behavior can be expressed in terms of a power law as well,

$$\tau \propto \xi^z \propto \varepsilon^{-\nu z}. \tag{4}$$

where ξ is the (divergent) correlation length, $\varepsilon = |1 - T/T_c|$, and z is the dynamic critical exponent.

In a finite system of linear size L, the divergence of the correlation length at criticality is limited by the system size. Exactly at T_c, the correlation length therefore saturates at $\xi \sim L$. Inserting this finite-size cutoff into the critical slowing-down relation $\tau \propto \xi^z$ gives

$$\tau_c \sim L^z. \tag{5}$$

Thus, at the critical temperature, the relaxation time is no longer controlled by the reduced temperature ε, but instead by the system size itself. As L increases, correlations require more time to propagate across the entire system, leading to a relaxation time that scales as a power law in L with exponent z.

A second relaxation time, the integrated relaxation time, is defined by the integral of the relaxation function

$$\tau_{\text{int}} = \int_0^\infty \phi(t)\, dt. \tag{6}$$

This quantity has particular importance for the determination of errors and is expected to diverge with the same dynamic exponent as the 'exponential' relaxation time.

1.2 Autocorrelation and integrated autocorrelation time for magnetization and energy

To quantify critical slowing down we compute the autocorrelation function of an observable A (for instance the magnetization M, the energy or the susceptibility):

$$C_A(t) = \langle A_s A_{s+t}\rangle - \langle A\rangle^2,$$

and the normalized autocorrelation (autocorrelation coefficient)

$$\rho_A(t) = \frac{C_A(t)}{C_A(0)}.$$

A convenient scalar measure of the memory of the Markov chain is the *integrated autocorrelation time*

$$\tau_{\text{int},A} = \frac{1}{2} + \sum_{t=1}^{T_{\max}} \rho_A(t),$$

where the sum is truncated at a cutoff $T_{\max}$ chosen e.g. by the window method (or until $\rho_A(t)$ has decayed to noise).

1.3 Extracting the dynamic exponent z

At criticality finite-size scaling for relaxational dynamics predicts that the characteristic relaxation time of the system at T_c scales with the linear size L as

$$\tau_c(L) \propto L^z,$$

because the correlation length is cut off by the system size $\xi \sim L$ and $\tau \propto \xi^z$. We estimate $\tau_c(L)$ by the integrated autocorrelation time of the magnetization (**explain why**), and then fit $\log \tau_c$ versus $\log L$ to extract z:

$$\log \tau_c = z \log L + \text{const.}$$

1.4 Susceptibility

For a system of $N = L^2$ spins at zero field is estimated from recorded magnetizations M_k with the FDT, Fluctuation-Dissipation Theorem:

$$\chi = \frac{1}{k_B T} \left(\langle M^2 \rangle - \langle M \rangle^2 \right)$$

The susceptibility is also defined thermodynamically as the derivative of the average magnetization with respect to an external magnetic field h: [2]

$$\chi = \left. \frac{\partial \langle M \rangle}{\partial h} \right|_{h=0}$$

In practice, this derivative is often approximated by a finite-difference scheme using a small nonzero field h_0:

$$\chi \approx \frac{\langle M \rangle_{+h_0} - \langle M \rangle_{-h_0}}{2h_0}$$

with matched burn-in across the $\pm h_0$ runs for each β

2 Algorithm

2.1 Metropolis Algorithm for the 2D Ising model

Metropolis method ([4]) configurations are generated from a previous state using a transition probability which depends on the energy difference between the initial and final states. The states produced follows a time ordered path, and this time is not a proper time but a Monte-Carlo time. For relaxational models, such as the stochastic Ising model, the time-dependent behavior is described by the master equation (1) :

$$\frac{\partial P_n(t)}{\partial t} = - \sum_{n \neq m} \left[P_n(t) W_{n \to m} - P_m(t) W_{m \to n} \right] \tag{7}$$

with the probability of the nth state occurring in a classical system is given by

$$P_n = \frac{e^{-\beta E_n}}{Z}$$

with Z the partition function which we don't know about. To bypass the epxression of Z, we generate a Markov chain of states, i.e. we generate each new state directly from the preceding state. If we produce the n_{th} state from the m_{th} state, and then calculate the relative probability which is the ratio of the individual probabilites, the denominator cancels. As a result, only the energy difference between the two states is needed : $\Delta E = E_n - E_m$

Indeed know thanks to the master equation (1) that in equilibrium :

$$P_n W_{n\to m} = P_m W_{n\to m}$$

Thus

$$\frac{Wn \to m}{Wm \to n} = \frac{P_m}{P_n} = e^{-\beta \Delta E}$$

Therefore,

$$W_{n\to m} = \begin{cases} \tau_0^{-1} e^{-\beta\Delta E} = e^{-\beta\Delta E} & \text{si } \Delta E > 0, \\ \tau_0^{-1} = 1 & \text{si } \Delta E \leq 0. \end{cases}$$

as we know that $W_{n\to m}$ is of the dimensional of 1/time and τ_0 is the time required to attempt a spin-flip that we choose equal to 1.

The steps of the algorithm are as follows:

1. Choose an initial state.
2. Choose a lattice site i.
3. Compute the energy change ΔE resulting from flipping the spin at site i.
4. Generate a random number r such that $0 < r < 1$.
5. If $r < e^{-\beta\Delta E}$, flip the spin.
6. Move to the next site and return to step (3).

2.1.1 Single-Spin Metropolis algorithm (checkerboard) on 2D Ising Model

In this work we use a *hot start* (random initial condition) for each run: the lattice spins are initialized independently to $+1$ or -1 with equal probability. The hot start avoids biasing the system toward a magnetized state and is particularly convenient when scanning temperatures around T_c.

We employ a single-spin *Metropolis* algorithm with a checkerboard (updating parity) scheme. The checkerboard scheme divides the square lattice into two sublattices (even/odd parity) so that all spins updated in the same micro-step have only neighbours belonging to the opposite parity; this allows an explicit, conflict-free update sequence:

$$\text{for parity} = 0, 1 : \quad \text{update all sites } (i,j) \text{ with } (i+j) \bmod 2 = \text{parity}.$$

Periodic boundary conditions are used throughout. For a proposed flip of spin $s_{i,j}$ the energy change is

$$\Delta E = 2 s_{i,j} \Big(J \sum_{\langle k \rangle} s_k + h \Big),$$

and the Metropolis acceptance probability is

$$P_{\text{accept}} = \min\big(1, e^{-\beta\Delta E}\big).$$

2.1.2 Monte Carlo time

The algorithm produces a time-ordered sequence of configurations; the index along this sequence is the *Monte Carlo time* (in units of sweeps when we define one sweep $= L \times L$ single-spin update attempts). Because this time is algorithmic rather than physical, we refer to relaxation times in units of Monte Carlo sweeps.

2.2 Wolff Algorithm for the 2D Ising Model

Wolff **(ref)** proposed an alternative algorithm based on the Fortuin–Kasteleyn theorem in which single clusters are grown and flipped sequentially. The algorithm begins with the random choice of a single site. Bonds are then drawn to all nearest neighbors which are in the same state with probability

$$p = 1 - e^{-K\delta_{\sigma_i \sigma_j}}$$

One then moves to all sites in turn which have been connected to the initial site and places bonds between them and any of their nearest neighbors which are in the same state with probability given by equation (6). The process continues until no new bonds are formed and the entire cluster of connected sites is then flipped. Another initial site is chosen and the process is then repeated.

A single Wolff cluster update consists of the following steps:

1. Randomly select a seed spin from the lattice.
2. Grow a cluster by adding neighboring spins with the same orientation with probability $p_{\text{add}} = 1 - \exp(-2\beta J)$, where J is the nearest-neighbor coupling and $\beta = 1/T$ is the inverse temperature.
3. Repeat the cluster growth iteratively for all new cluster members until no further spins are added.
4. Flip all spins in the cluster simultaneously.

Depth-first search is used to traverse and grow the cluster efficiently. DFS avoids revisiting the same spins multiple times, making the algorithm fast even for very large clusters.

Assumptions in our code:

- The initial configuration is the same that we did for Metropolis i.e. hot start : random, with spins sampled uniformly from $\{-1, +1\}$.

- Periodic boundary conditions are applied in both directions, allowing each spin to interact with four nearest neighbors regardless of its position on the lattice.

- Interactions are limited to nearest neighbors with uniform coupling strength J.

- The algorithm assumes ergodicity and sufficient equilibration to sample the canonical ensemble.

2.2.1 Monte Carlo time, equilibration and measurements

The Monte Carlo time was rescaled for the Wolff algorithm using the mean number of lipped clusters $\langle C \rangle$ per step:

$$\Delta t = \frac{\langle C \rangle}{L^2}, \quad t_{\text{MC}} = \sum \Delta t \tag{8}$$

This makes the Wolff steps comparable to a full Metropolis sweep.

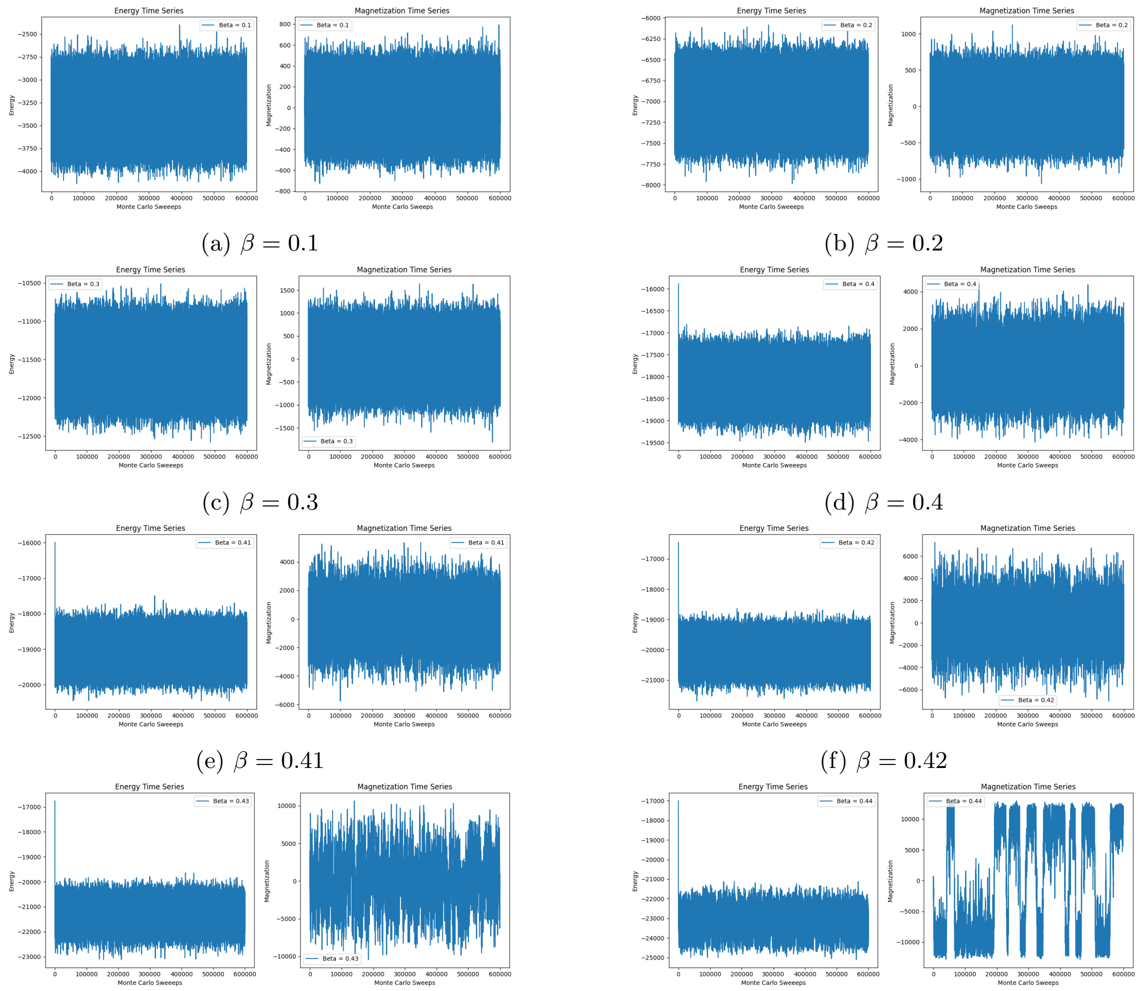
Time series of energy E, magnetization M, absolute magnetization $|M|$, and squared magnetization M^2 were analyzed for each β to study fluctuations and equilibrium behavior.

3 Numerical Analysis and Results

3.0.1 Energy and Magnetization time series

Metropolis Before recording measurements we perform an equilibration phase of several thousand sweeps (or more near T_c) to ensure the system has reached stationarity. After equilibration we record observables (energy, total magnetization) at regular intervals. The choice of recording interval should be made in view of the autocorrelation time (below) to reduce correlation between successive measurements.

Plotting the energy and magnetization time series for various inverse temperatures β allows us to verify that the system reaches equilibrium, identify enhanced fluctuations near the critical temperature β_c, and observe critical slowing down. These plots are also used to determine the appropriate *burn-in* period, i.e., the number of initial sweeps to discard before measurements, and to choose suitable recording intervals between successive measurements, ensuring that the collected data are sufficiently decorrelated.



(a) $\beta = 0.1$ (b) $\beta = 0.2$

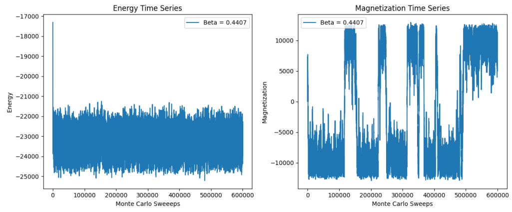
(c) $\beta = 0.3$ (d) $\beta = 0.4$

(e) $\beta = 0.41$ (f) $\beta = 0.42$

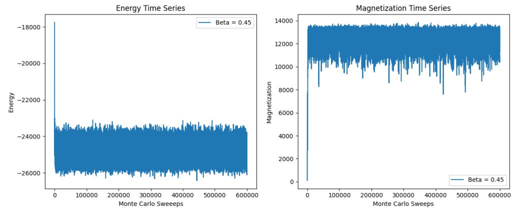
(g) $\beta = 0.43$ (h) $\beta = 0.44$

Figure 1: Magnetization and Energy Time Series for $\beta = 0.1$ to 0.44.

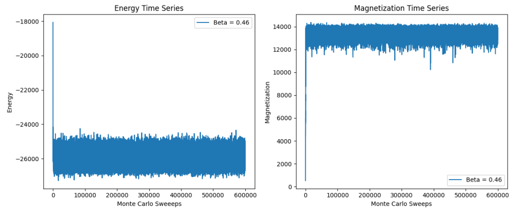
At low values of β (high temperatures), both energy and magnetization fluctuate around mean values with relatively small amplitudes, indicating that the system is in a disordered phase and has reached equilibrium quickly. As β approaches the critical region near $\beta_c \approx 0.44$, the magnetization oscillates near zero, indicating a disordered, paramagnetic phase.



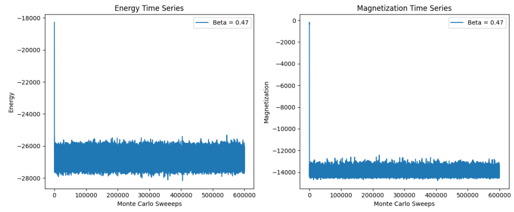
(a) $\beta = 0.4407$



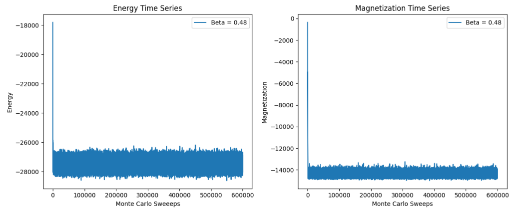
(b) $\beta = 0.45$



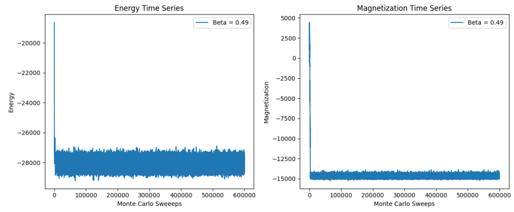
(c) $\beta = 0.46$



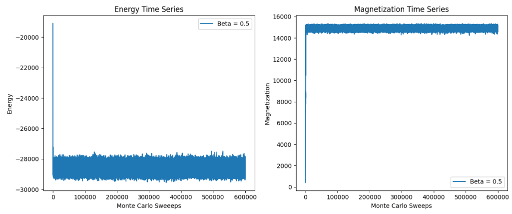
(d) $\beta = 0.47$



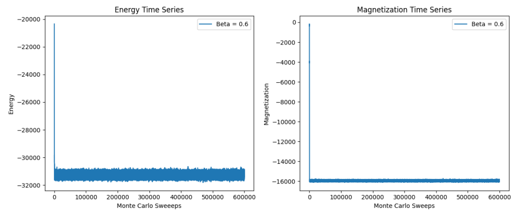
(e) $\beta = 0.48$



(f) $\beta = 0.49$



(g) $\beta = 0.5$



(h) $\beta = 0.6$

Figure 2: Magnetization and Energy Time Series for $\beta = 0.4407$ to 0.6.

At the critical point $\beta = 0.4407$, the energy fluctuates around higher values and the magnetization oscillates near zero, indicating a disordered, paramagnetic phase. As β increases, the energy quickly stabilizes at lower values and the magnetization converges to a non-zero value, reflecting the onset of an ordered, ferromagnetic phase. Near the critical region (e.g., $\beta \approx 0.45$–0.48), magnetization exhibits abrupt flips, characteristic of critical slowing down. These series illustrate the expected transition from disorder to order in the Ising model under Metropolis dynamics. These plots also illustrate critical slowing down, as configurations near β_c require longer Monte Carlo times to decorrelate, highlighting the need for a sufficiently long burn-in and appropriately spaced measurements.

Wolff

3.1 Susceptibility

Metropolis We thus plot with the FDT theorem the susceptibility as function of β.

Then we can compare the value of the susceptibility with the FDT and with a small non zero external field.

3.1.1 Wolff

Magnetic susceptibility χ is computed using the fluctuation-dissipation theorem:

$$\chi = \beta \left(\langle |M|^2 \rangle - \langle |M| \rangle^2 \right) , \tag{9}$$

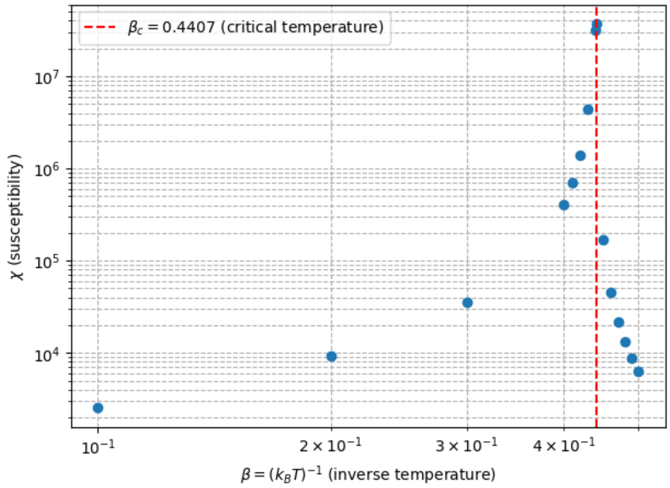


Figure 3: Susceptibility as a function of β. The red dashed line indicates the critical value β_c.

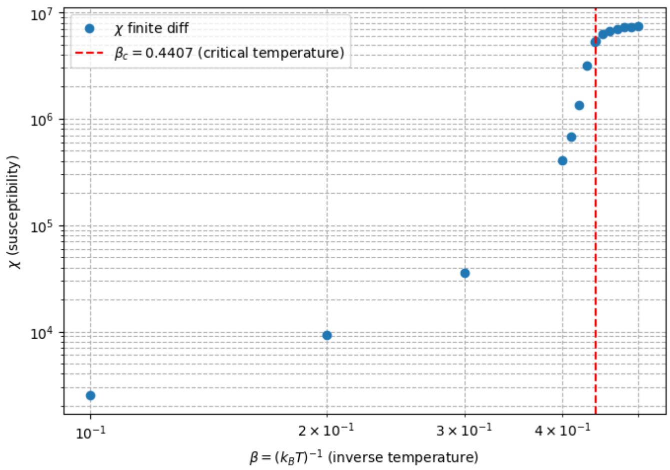


Figure 4: Susceptibility as a function of β with small external field. The red dashed line indicates the critical value β_c.

where M is the total magnetization and $\beta = 1/(k_B T)$. The absolute value of M is used because, in the absence of an external magnetic field ($H = 0$), the system is symmetric under spin inversion ($M \leftrightarrow -M$). As a result, the thermal average $\langle M \rangle$ vanishes even below the critical temperature T_c, despite the presence of spontaneous magnetization. Taking the absolute value ensures that the measured fluctuations reflect the true magnetic behavior of the system, avoiding artificial cancellation due to symmetry. Hence, $\langle |M| \rangle$ provides a physically meaningful estimate of the net magnetization and its fluctuations.

3.2 Autocorrelation and integrated autocorrelation time for magnetization and energy

3.3 Dynamical critical exponent

For each lattice size L, we load long Monte Carlo time series of the magnetization and energy at the critical inverse temperature $\beta_c = 0.4407$, that we obtained previously. We compute the normalized autocorrelation function using an FFT-based estimator, and determine the integrated autocorrelation time τ_{int} using Sokal's self-consistent windowing procedure. An iterative scheme is applied to automatically estimate and remove the thermalization (burn-in) period, ensuring that the remaining data satisfy $N \gg 50\,\tau_{\text{int}}$. The resulting autocorrelation times for each L are then fitted to the expected dynamic scaling form $\tau_{\text{int}}(L) \sim L^z$ in log–log scale, allowing us to extract the dynamic critical exponent z for both the magnetization and the energy.

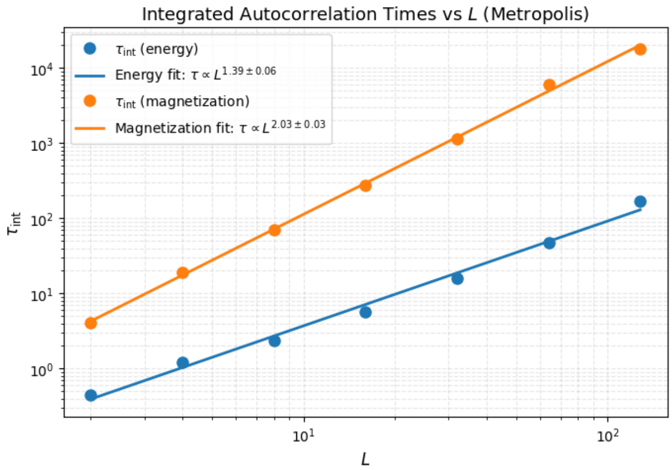


Figure 5: Integrated autocorrelation time as a function of L. The red dashed line indicates the critical value β_c.

4 Comparison of algorithm efficiency at criticality

References

[1] A Guide to Monte Carlo Simulations in Statistical Physics, – David P. Landau, Kurt Binder – 2nd ed, Cambridge, UK, New York, England, 2005

[2] COMPLEXITY AND CRITICALITY (Imperial College Press Advanced – Kim Christensen, Nicholas R. Moloney – Imperial College Press advanced physics texts

[3] "Monte Carlo Methods in Statistical Mechanics: Foundations and New Algorithms", Sokal (1997).

[4] Metropolis et al., 1953 – "Equation of State Calculations by Fast Computing Machines"